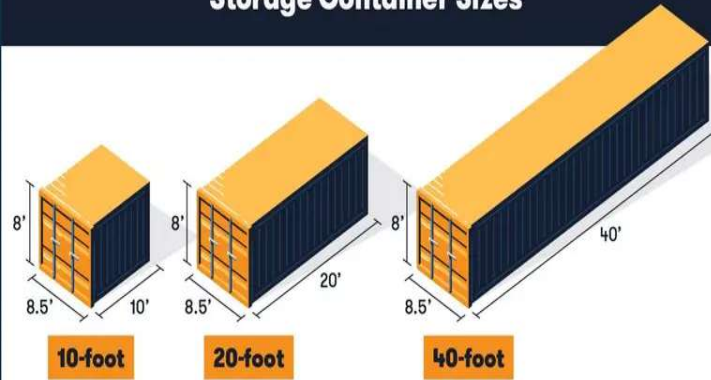


Introduction To Docker

01

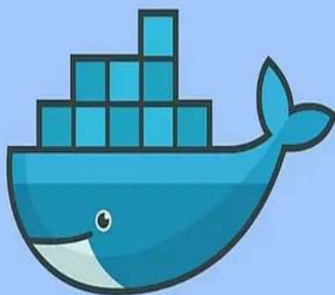
What is Docker, Why use Docker,
Docker Architecture Overview

Storage Container Sizes

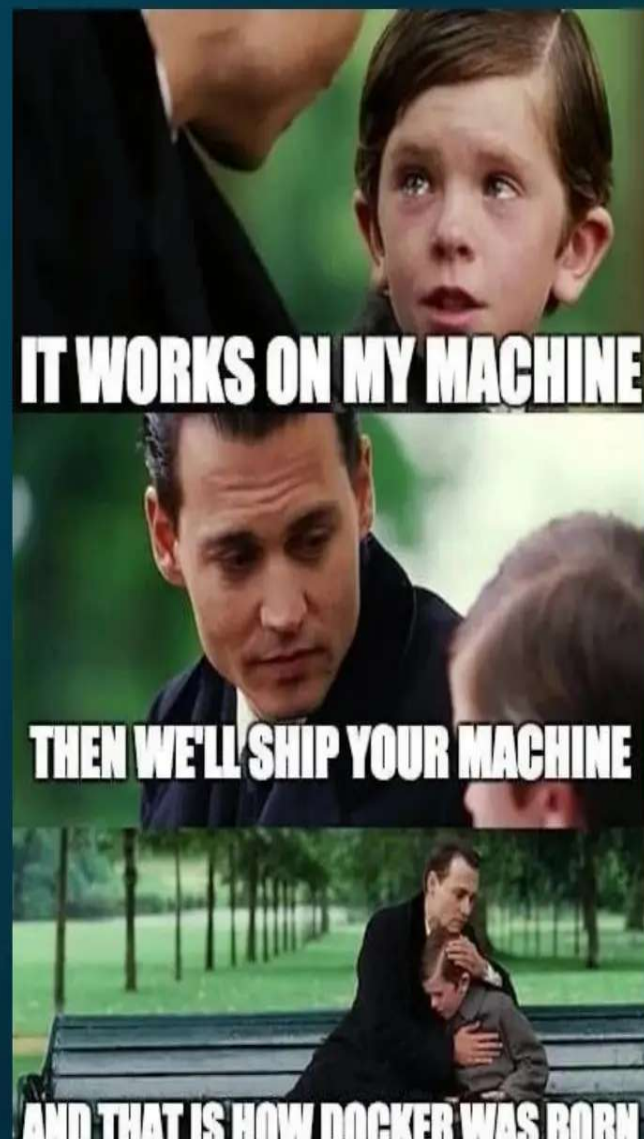




What is Docker?



- ❖ Docker is an open-source **platform used for containerization**, allowing you to package an application and its dependencies into a standardized unit called a container.
- ❖ Containers are lightweight, portable, and isolated environments that ensure consistent behavior across different computing environments.



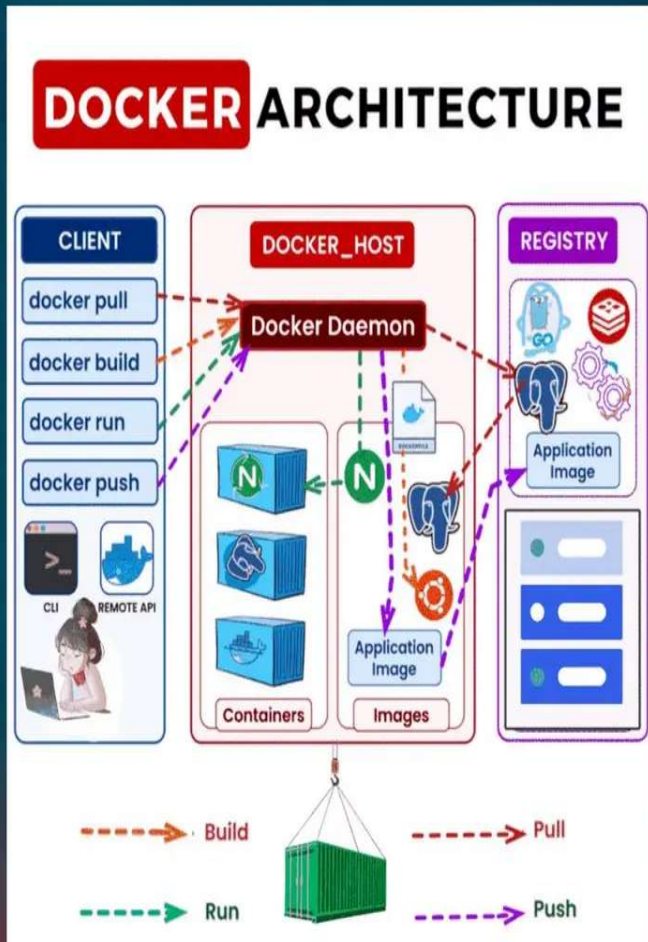


Why use Docker?



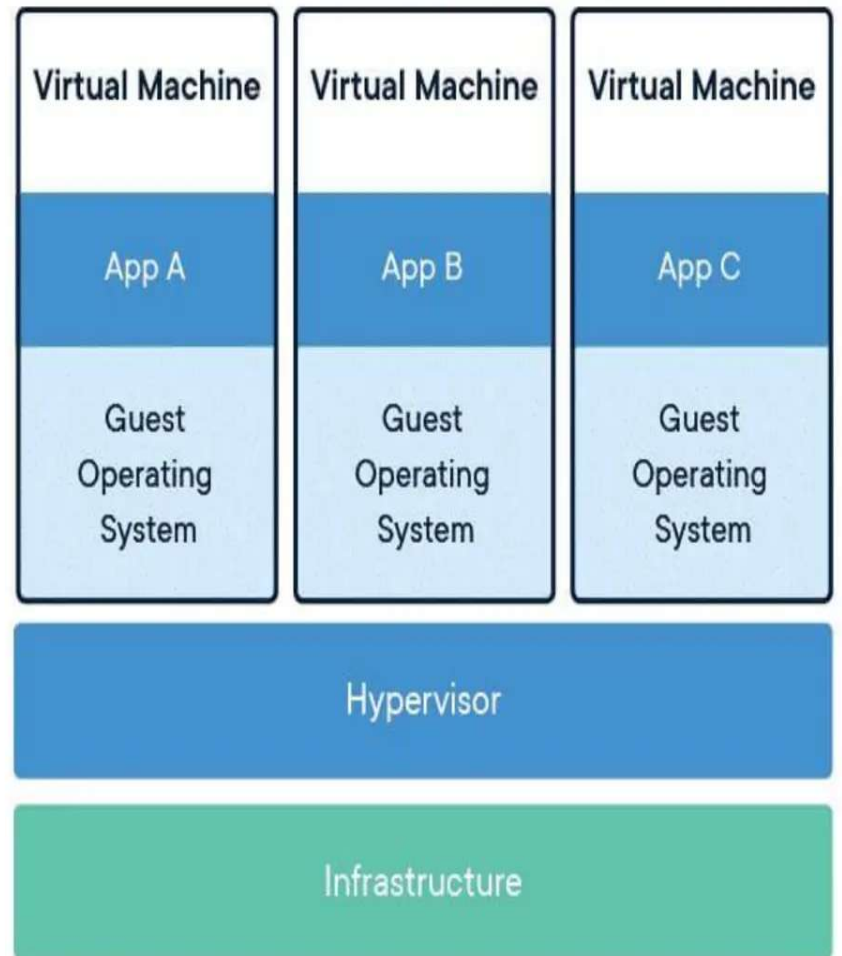
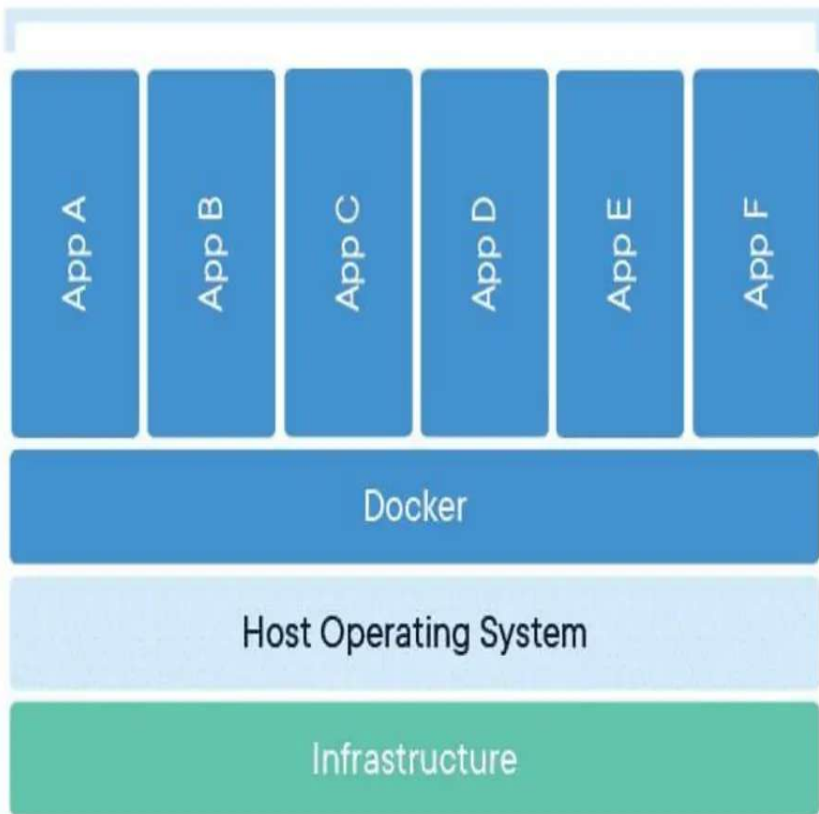
- ❖ Docker simplifies the process of developing, shipping, and deploying applications by encapsulating them in containers.
- ❖ Benefits of Docker include faster application deployment, resource efficiency (compared to virtual machines), improved scalability, and simplified DevOps workflows.

Docker Architecture Overview:



- ❖ Docker Engine: The core component of Docker that runs and manages containers.
- ❖ Docker Client: CLI tool used to interact with Docker Engine.
- ❖ Docker Daemon: Background service responsible for managing containers, images, networks, and volumes.
- ❖ Docker Registry: Stores Docker images (e.g., Docker Hub, private registries).

Containerized Applications



Advantages of Proposed System

02

○ Portability

○ Consistency

○ Efficiency

○ Scalability



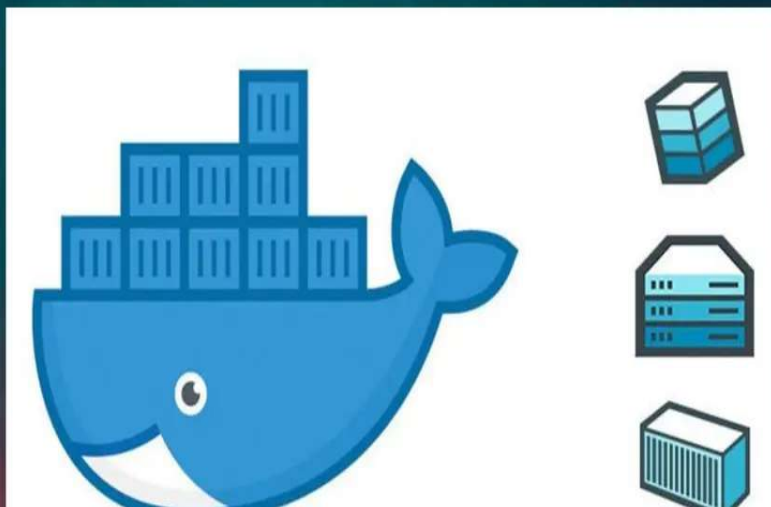
Portability: Docker containers encapsulate applications and their dependencies into portable units, making them highly portable across different environments, from development to testing and production. This portability ensures consistent behavior and eliminates "works on my machine" issues.



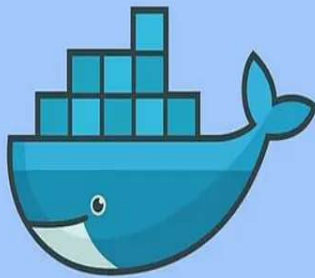
Efficiency: Docker containers are lightweight and share the host OS kernel, leading to efficient resource utilization. They have minimal overhead compared to virtual machines (VMs), allowing for higher density deployments and reduced infrastructure costs.



Isolation: Docker provides process-level isolation, ensuring that applications run independently of each other. This isolation prevents conflicts between applications and enables safer experimentation with different



DOCKER



Consistency: With Docker, developers can create Dockerfiles to define the environment and dependencies required for their applications. This ensures consistent development, testing, and production environments, reducing deployment errors and streamlining workflows.



Scalability: Docker's containerization model facilitates horizontal scaling by replicating containers across multiple nodes or instances. This scalability enables applications to handle varying workload demands and ensures optimal performance during peak usage times.



Speed: Docker containers start up quickly and have faster deployment times compared to traditional deployment methods. This speed is especially beneficial for continuous integration/continuous deployment (CI/CD) pipelines, where rapid iteration and deployment are required.



Disadvantages of Proposed System

03

○ Portability

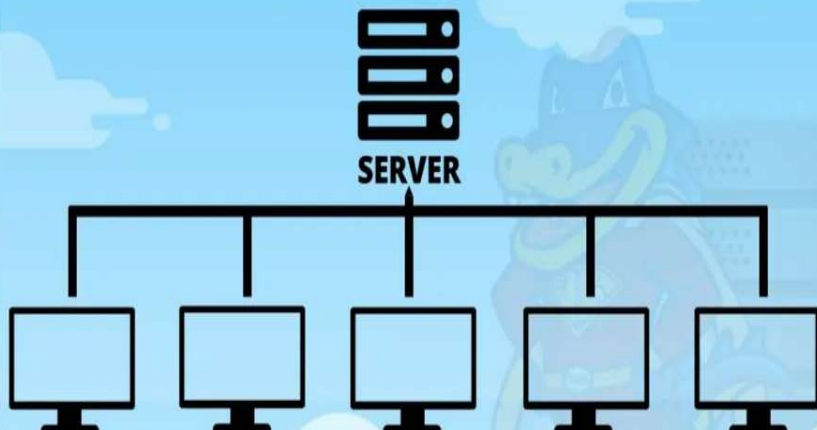
○ Consistency

○ Efficiency

○ Scalability



SHARED HOSTING



Complexity: Docker introduces additional complexity, especially for users who are new to containerization concepts. Understanding Dockerfile syntax, image creation, container orchestration, networking, and storage management can be challenging for beginners.



Security Concerns: While Docker provides isolation at the process level, it is still possible for vulnerabilities or misconfigurations to compromise container security. Shared kernel vulnerabilities, improper container configurations, and insecure container images can pose security risks.



Networking Complexity: Docker's networking capabilities, while powerful, can be complex to configure and manage, especially in distributed or multi-container environments. Issues such as container-to-container communication, network

Results and Conclusion

03

Results and Conclusion



- ❖ In conclusion, the adoption of Docker containers has significantly transformed our software development and deployment practices, leading to tangible benefits and improvements across various aspects of our projects.
- ❖ Docker's portability, scalability, streamlined workflows, resource efficiency, consistent environments, security features, and deployment flexibility have made it an invaluable tool in our development toolkit.

Litrature Survey

03

1) Base Systems for Docker Containers - Security Analysis (September 2022):

Docker based containerization is currently one of the most popular methods of delivering and creating of a software. It allow multiple teams to standarize their work, but also to reduce disadvantages of virtual machines that can impact performance and usability. This work concerns security of base systems, focusing on distroless. Base container images are one of the critical parts of a cloud environment. The results of analysis presented here allow for independent and objective comparison of advantages and disadvantages of various containers' base systems which are widely used in orchestration platforms such as Kubernetes and OpenShift

2) Research on Using Docker Container Technology to Realize Rapid Deployment Environment on Virtual Machine (September 2022):

At present, with the rapid development of IT technology, the traditional computing mode has become more and more difficult to meet the processing and various engineering or scientific computing tasks. In fact, with the gradual popularization of computers and the continuous progress of semiconductor technology, the computing model has experienced several major changes. These changes can be summarized into four stages: character dumb terminal host, client server, cluster computing and cloud computing. Virtual machine services account for a large proportion of cloud computing services. The implementation of virtual machine can enable users to improve use efficiency, reduce use cost, facilitate migration and backup, and reduce maintenance cost. Docker container technology has the above advantages at the same time. How to combine the two to improve the operation and maintenance efficiency and deploy the project environment faster.

<https://docs.docker.com/>

References

03

- 1.Arkadiusz Maruszczak,Michal WalkowskiSlawomir Sujecki, “Base Systems for Docker Containers - Security Analysis” (September 2022)
- 2.Wei Wang,”Research on Using Docker Container Technology to Realize Rapid Deployment Environment on Virtual Machine”(September 2022)
- 3.Li You¹and Hui Sun,”Research and Design of Docker Technology Based Authority Management System”(2022)

<https://docs.docker.com/>

THANKS!

Do you have any questions?



shreyasprabhu26



ShreyasPrabhu26